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Information Systems
Labwork 5

`employee` table

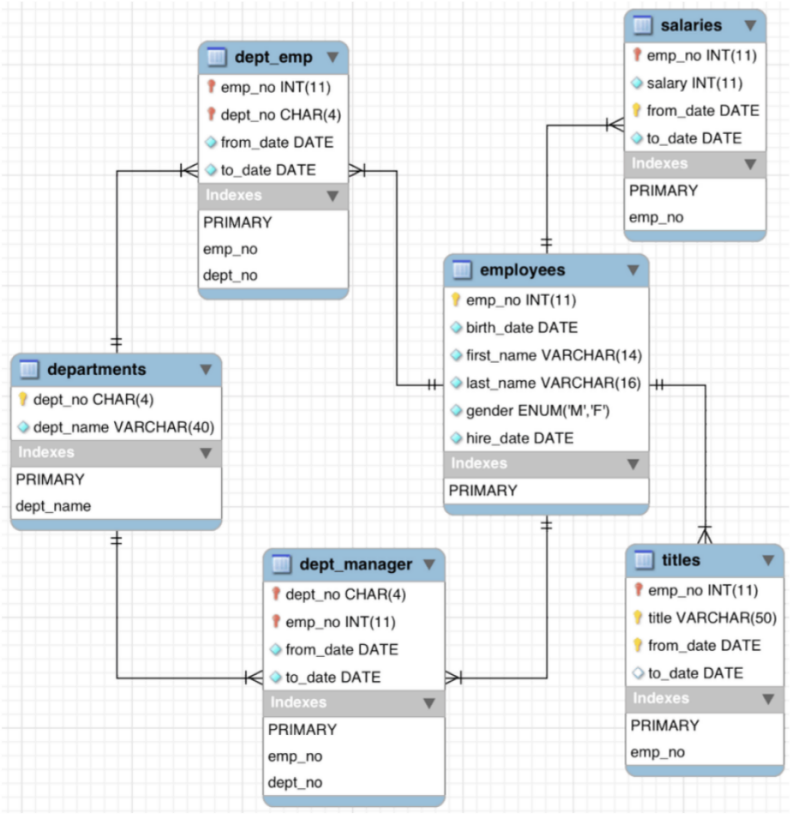


Figure 1: `employees` database schema

1NF. Every column holds a single atomic value (no lists, no comma-separated values), and no table has repeated columns. In particular, histories are stored as *rows*, not as columns: an employee's salaries are many rows of `salaries`, not columns `salary1`, `salary2`, ... Every table has a primary key, so no two rows are identical.

2NF. A partial dependency is only possible when the primary key is composite, so:

- `employees` (key `emp_no`) and `departments` (key `dept_no`) have single-column keys and are in 2NF automatically.
- `dept_emp` and `dept_manager` each have one composite primary key (`emp_no`, `dept_no`), both columns also being foreign keys. `from_date` and `to_date` need the whole key: `emp_no` alone is not enough (an employee who moved has a different period in each department), and neither is `dept_no` (a department has many employees with different periods).

- `titles` has the composite key (`emp_no`, `title`, `from_date`). `to_date` needs all three: one employee holds several titles, one title is held by many employees, and `from_date` distinguishes two separate periods with the same title.
- `salaries` has the composite key (`emp_no`, `from_date`). `salary` and `to_date` need both: `emp_no` alone is not enough (an employee gets a new salary every year), and neither is `from_date` (many employees get a raise on the same day with different amounts).

3NF. No non-key column determines another non-key column:

- `employees`: none of `birth_date`, `first_name`, `last_name`, `gender`, `hire_date` determines another. For example, `first_name` $\not\to$ `gender`: two employees can share a first name and have different genders.
- `departments`: the only non-key column, `dept_name`, is `UNIQUE`, so it is a second candidate key, not a transitive dependency.
- `dept_emp`, `dept_manager`: between `from_date` and `to_date`, neither determines the other (the length of a period varies).
- `titles` has a single non-key column (`to_date`), so it cannot have a transitive dependency.
- `salaries`: `salary` $\not\to$ `to_date` and `to_date` $\not\to$ `salary`; the same amount can be paid over different periods.

Table	**PK**	**1NF: Columns with atomic value, no list in column**	**2NF: Is there any composite key?**	**3NF: Are there any non-key → non-key?**
`employees`	`emp_no`	`birth_date`, `first_name`, `last_name`, `gender`, `hire_date`	None, only `emp_no`	None, all attributes depend on `emp_no`
`departments`	`dept_no`	`dept_name`	None, only `dept_no`	None, only one attribute and one primary key
`dept_emp`	`emp_no` `dept_no`	`from_date`, `to_date`	1 composite PK `dept_no` and `emp_no`, both are primary keys and both are foreign keys too	None, `from_date` and `to_date` are independent
`dept_manager`	`emp_no` `dept_no`	`from_date`, `to_date`	Same as `dept_emp`	Same as `dept_emp`
`titles`	`emp_no` `title` `from_date`	`to_date`	1 composite PK `emp_no`, `title` and `from_date` with `emp_no` is also a foreign key	None, `to_date` depends on the entire composite key
`salaries`	`emp_no` `from_date`	`salary`, `to_date`	1 composite key of `emp_no` and `from_date` with `emp_no` is also a foreign key	None, `salary` and `to_date` requires the entire composite key

The `employee` database satisfies third normal form (3NF).

1. In all tables, each column contains atomic value and there are no repeated columns within a table → `employee` satisfies first normal form (1NF)
2. All non-key column depends on the primary keys:
   - `employees` table: all attributes are identified by the primary key `emp_no`.
   - `titles` table: this table has a composite primary key. It consists of `emp_no`, `title` and `from_date`. Of which, `emp_no` is also a foreign key linking `titles` to `employees`.
   - `departments` table: this table contains a primary key of `dept_no` and attribute `dept_name` which is unique and depends on the primary key.

- `dept_emp` and `dept_manager` both have two primary key which happen to be foreign key. `emp_no` links back to table `employees` and `dept_no` links back to `departments`.

3. It has no transitive dependencies, all the non-key columns depends on the key column.

Order form

Normalize the following document to 3NF

Order Form			
Order number	on7126		
Order date	11/07/2016		
Customer name	Tran Giang Son		
Customer phone	+84-4 37 91 69 60		
Customer address	18 Hoang Quoc Viet		
City	Hanoi, Vietnam		
ProdId	Description	Quantity	Price
P101	Intel i7 6700K	1	$349
P102	ASUS Z170A-P	1	$149
P103	Corsair DDR4 4 × 8GB	1	$469
P104	Samsung EVO 850 512GB	1	$249
P105	GeForce GTX 1080 8G	1	$725
P106	Corsair CS750M 80+ Gold	1	$119
P107	Corsair Carbide 400R Mid	1	$85

Order database schema normalized to 3NF

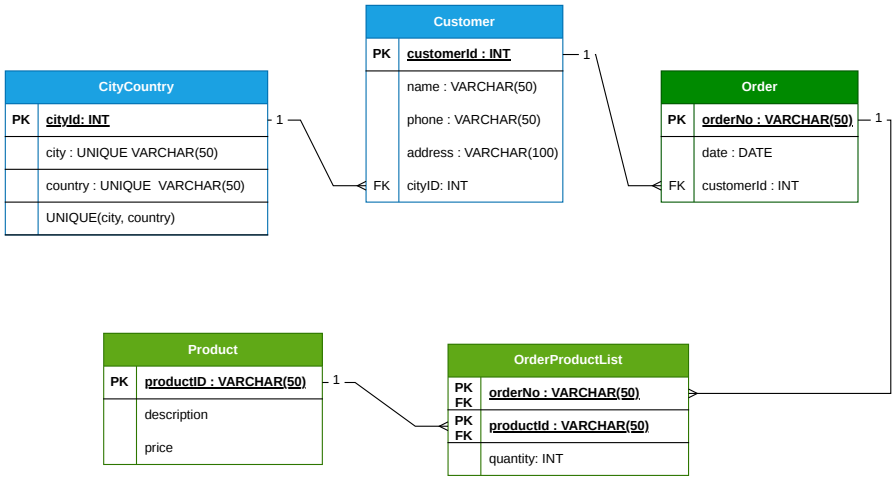


Figure 2: Order database schema